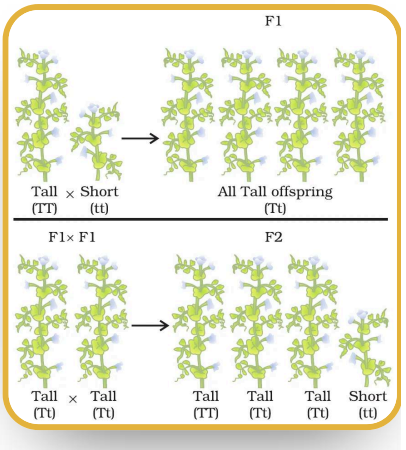
MT Classes, Motihari

Heredity

Class 10 Science, Chapter 8 (NCERT, CBSE 2026-27)

Variation, Mendel's pea experiments, dominant and recessive traits, the 3 : 1 and 9 : 3 : 3 : 1 ratios, genes and chromosomes, and how sex is decided in humans, explained beside the textbook's own diagrams, with a Punnett square builder, real CBSE board questions with answers, every NCERT question solved and a quiz.



📌 Chapter snapshot

- Chapter 8 of NCERT Class 10 Science (reprint 2026-27), in Unit II “World of Living”, which carries **25 marks** in the 80-mark theory paper together with Life Processes, Control and Coordination, and How do Organisms Reproduce?
- It is a short chapter (6 pages), but it is asked **every year**: a 4-mark **case study on Mendel** (2023, 2024, 2026), MCQs on **sex determination** and the **dihybrid F1**, and a 3-mark question on **chromosomes and DNA stability** (2025).
- Learn by heart: **3 : 1** (F2 phenotype, one trait), **1 : 2 : 1** (F2 genotype), **9 : 3 : 3 : 1** (F2 phenotype, two traits), **XX** girl, **XY** boy, sex decided by the **father**.

What is this chapter about?

Reproduction makes new individuals that are **similar but subtly different**. This chapter is about **how those differences (variations) are created and inherited**.

- Some variation arises even in **asexual** reproduction (small errors in DNA copying).
- **Sexual** reproduction makes the number of successful variations much larger.
- **NCERT example:** a field of **sugarcane** (grown vegetatively) shows very little variation among plants. But in **human beings** and other sexually reproducing animals, quite distinct variations are visible.



Chapter opener: all living things are branches of one family tree

🧩 Key words first

Heredity: the transmission of characters (traits) from parents to their offspring.
Variation: the differences among individuals of the same species, or between parents and their offspring.
Trait: a characteristic, such as height of a plant, colour of a seed, or type of earlobe.

8.1 Accumulation of variation during reproduction

- Inheritance from the previous generation gives the next generation **both** a common basic body design **and** subtle changes in it.
- When this new generation reproduces, the second generation has differences **inherited from the first**, plus **newly created** differences. So variations **accumulate** (pile up) over generations (Fig. 8.1).
- Fig. 8.1 shows a **single individual** reproducing, as in **asexual** reproduction. If one bacterium divides, and the two divide again, the four bacteria are **very similar**, with only minor differences from small inaccuracies in DNA copying.
- With **sexual** reproduction, **even greater diversity** is generated, because DNA from two parents is combined.

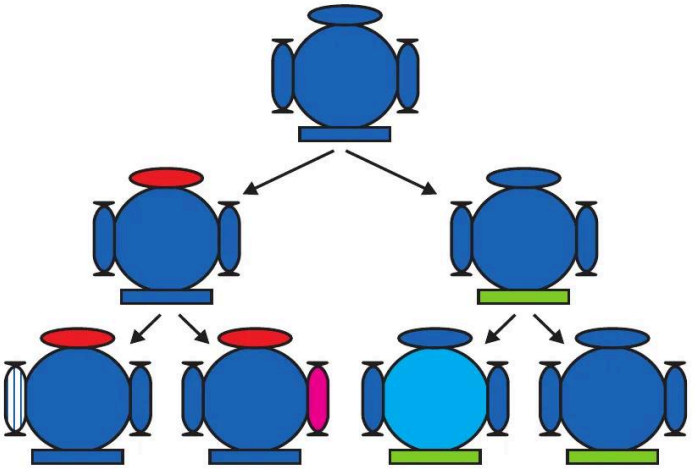


Fig. 8.1 Creation of diversity over succeeding generations. Each of the four in the bottom row is different: some differences are new, others are inherited from their parents.

Do all variations survive equally?

- **No.** Depending on the nature of the variation, different individuals get **different kinds of advantages**.
- **NCERT example:** bacteria that can withstand heat survive better in a **heat wave**.
- **Selection of variants by environmental factors** forms the basis for **evolutionary processes**.

> ⭐ **Exam line**
>
> Variations help a species survive: if the environment changes, the individuals carrying a useful variation survive and reproduce, so the species does not die out. Sexual reproduction produces **more** variation than asexual reproduction.

8.2 Heredity and inherited traits

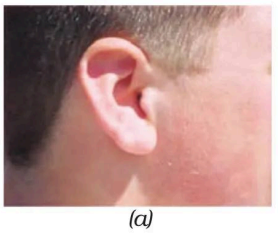
The most obvious outcome of reproduction is still individuals of **similar design**. The **rules of heredity** decide how traits and characteristics are **reliably inherited**.

8.2.1 Inherited traits

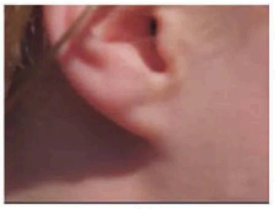
- A child bears **all the basic features of a human being**.
- But it does **not look exactly like its parents**, and human populations show a great deal of variation.
- **Example:** the earlobe. In some of us the lowest part of the ear is **free** (hangs loose); in others it is **attached** closely to the side of the head (Fig. 8.2). These are two variants found in human populations.

– **Activity 8.1: Survey earlobes in your class and in parents. What rule can you suggest?**

Count students with free and attached earlobes and work out percentages. Then compare each student with their parents. You will usually find that two parents with **attached** lobes have children with attached lobes, while parents with **free** lobes can have children of either type. This suggests that **free earlobe behaves as dominant** and **attached earlobe as recessive**: a child shows the attached type only if it gets the "attached" version from **both** parents. (Real earlobe shape is influenced by more than one gene, so expect a few exceptions.)



(a)



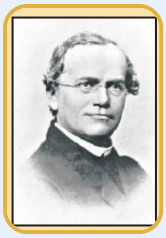
(b)

Fig. 8.2 (a) Free and (b) attached earlobes

8.2.2

Rules for inheritance: Mendel's contributions

- Both the **father and the mother contribute practically equal amounts of genetic material** to the child.
- So each trait can be influenced by both paternal and maternal DNA: for **each trait there are two versions** in every child. Which one shows? Mendel worked out the rules.



Gregor Johann Mendel (1822–1884)

Educated in a monastery, he studied science and mathematics at the **University of Vienna**. He failed the exams for a teaching certificate, went back to his monastery and started growing **garden peas**. Others had studied inheritance before, but Mendel blended science with mathematics and was the **first to keep count** of individuals showing a trait in each generation. This led him to the **laws of inheritance**. He is called the **Father of Genetics**.

Why garden peas? (extra, often asked)

Clear contrasting traits
Round/wrinkled seeds, tall/short plants, violet/white flowers, yellow/green seeds.

Self-pollinating
Pure lines are easy to keep, and crosses can be made by hand.

Quick and many
Short life cycle and many seeds per plant, so large counts are possible.

The monohybrid cross (one trait: height)

1. **Parents (P):** a pure **tall** plant (TT) is crossed with a pure **short** plant (tt).
2. **F1 (first generation): all plants were tall**. There were **no halfway** (medium-height) plants. Only one parental trait was seen, not a mixture.
3. Were these F1 tall plants the same as the tall parent? To test this, both the parental tall plants and the F1 tall plants were allowed to **self-pollinate**.
4. Parental tall plants gave **all tall** progeny. But the **F2** progeny of the F1 tall plants were **not all tall**: **one quarter were short**.
5. **Conclusion:** both tallness and shortness were inherited in the F1 plants, but **only tallness was expressed**.

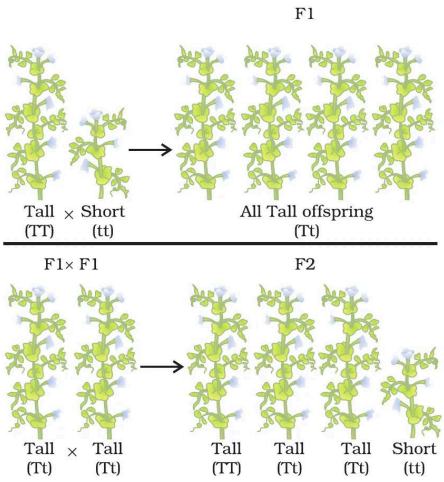


Fig. 8.3 Inheritance of traits over two generations

What Mendel proposed

- Two copies of a **factor** (now called a **gene**) controlling each trait are present in a sexually reproducing organism.
- The two copies may be **identical** (TT or tt) or **different** (Tt), depending on the parentage.
- Both **TT and Tt are tall**; only **tt is short**. A single copy of 'T' is enough to make the plant tall; both copies must be 't' for it to be short.
- A trait like 'T' that shows even with one copy is **dominant**. A trait like 't' that shows only when both copies are the same is **recessive**.

Term	Meaning	Example
Gene	A section of DNA that provides information for one protein; the unit of heredity	Gene for height
Alleles (extra)	Different forms of the same gene	T and t
Genotype (extra)	The genetic make-up for a trait	TT, Tt, tt
Phenotype (extra)	The trait that is seen	Tall, short
Homozygous / pure	Both copies the same	TT, tt
Heterozygous / hybrid	Two different copies	Tt
F1, F2	First and second filial (offspring) generations	F1 = Tt; F2 = TT, Tt, tt

Writing the cross (how to answer in the board exam)

F1 × F1: Tt × Tt		
Gametes →	**T**	**t**
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (short)

> ⭐ **Ratios to remember (monohybrid)**
>
> F2 **phenotypic ratio** = Tall : Short = **3 : 1**
>
> F2 **genotypic ratio** = TT : Tt : tt = **1 : 2 : 1**

Activity 8.2: What experiment would confirm that the F2 generation really had a 1 : 2 : 1 ratio of TT, Tt and tt?

Let **every F2 plant self-pollinate** and grow its seeds (the F3). The short F2 plants (tt) give **only short** plants. Of the tall F2 plants, about **one-third** give **only tall** plants (they are TT), and about **two-thirds** give **tall and short in 3 : 1** (they are Tt). Counting these gives TT : Tt : tt = 1 : 2 : 1. (Crossing each tall F2 plant with a short tt plant works too: TT gives all tall, Tt gives half tall and half short.)

Fig. 8.4: which trait is dominant?

The book asks you to work this out. Parents: **violet** flower × **white** flower → F1 is **violet** → F1 self-fertilised → F2 has **3 violet : 1 white**.

Answer: violet flower colour is **dominant** (it alone shows in F1) and **white** is **recessive** (it hides in F1 and reappears in one-quarter of F2).

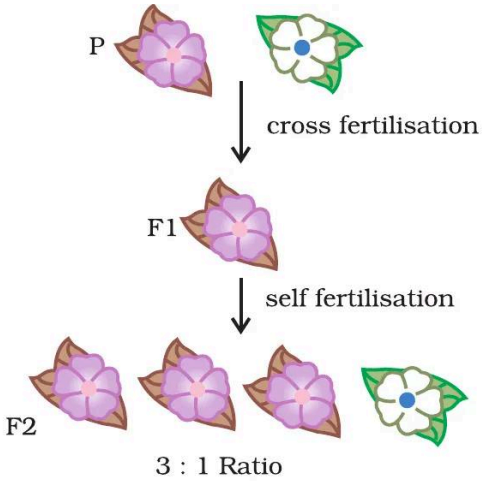


Fig. 8.4 Violet × white: 3 : 1 in F2

Tool

Punnett square builder

Pick the genotype of each parent. The square fills itself and gives the ratios. Use it to check any cross in your homework.

8.2.2 Two traits at once: independent inheritance

What happens when plants differing in **two** characters are crossed?

- **Tall plant with round seeds × short plant with wrinkled seeds** → F1 are **all tall with round seeds**. So tallness and round seeds are **dominant**.
- F1 plants self-pollinated → F2 has tall-round and short-wrinkled plants (parental types), **and also new combinations**: **tall-wrinkled** and **short-round**.
- So the tall/short trait and the round/wrinkled trait are **independently inherited**.

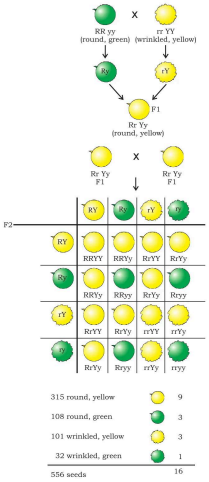


Fig. 8.5 Independent inheritance of two separate traits, shape and colour of seeds

The cross in Fig. 8.5 (seed shape and colour)

1. **P:** RRyy (round, green) × rrYY (wrinkled, yellow). Gametes: **Ry** and **rY**.
2. **F1:** all **RrYy (round, yellow)**. Round and yellow are dominant.
3. Each F1 plant makes **four kinds** of gametes in equal numbers: **RY, Ry, rY, ry**.
4. **F2:** 4 × 4 = 16 combinations in the checkerboard.

F2 seeds	Count (Fig. 8.5)	Ratio	Type
Round, yellow	315	**9**	Parental (F1 type)
Round, green	108	**3**	Parental (P1 type)
Wrinkled, yellow	101	**3**	Parental (P2 type)
Wrinkled, green	32	**1**	**New combination**
Total	**556**	**16**	

💡 Quick check

556 ÷ 16 ≈ 34.75. Then 9 × 34.75 ≈ 313, 3 × 34.75 ≈ 104, 1 × 34.75 ≈ 35: very close to Mendel's counts. Also note: round : wrinkled = (315 + 108) : (101 + 32) = 423 : 133 ≈ **3 : 1**, and yellow : green = 416 : 140 ≈ **3 : 1**. Each trait still follows 3 : 1 on its own, which is exactly what "independent" means.

⭐ Exam line

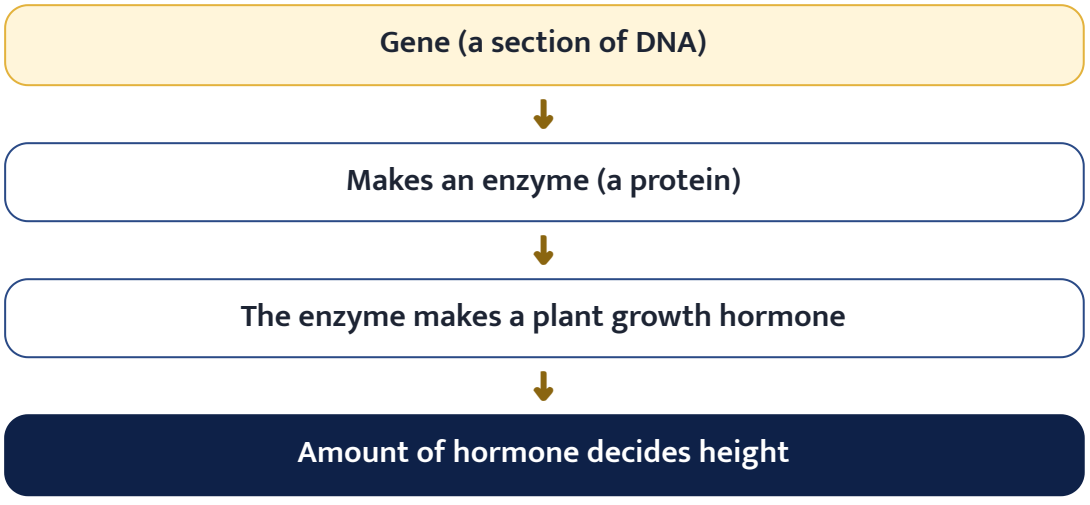
In a dihybrid cross the F2 **phenotypic ratio is 9 : 3 : 3 : 1**. The appearance of **new combinations** of traits (like round-green and wrinkled-yellow when the parents were round-green and wrinkled-yellow, or tall-wrinkled when the parents were tall-round and short-wrinkled) shows that traits are **inherited independently**.

8.2.3

How do these traits get expressed?

- Cellular **DNA** is the information source for making **proteins**.
- A section of DNA that provides information for **one protein** is called the **gene** for that protein.

NCERT example: how a gene controls plant height



Enzyme works efficiently

Lots of hormone is made

The plant is

tall

.

Gene altered, enzyme less efficient

Less hormone is made

The plant is

short

.

Thus **genes control characteristics, or traits**.

Why each germ cell has only one set of genes

- If both parents help decide a trait, **both must contribute a copy of the same gene**. So each pea plant has **two sets of all genes**, one from each parent.
- For this to work, each **germ cell must have only one gene set**.
- If a germ cell simply passed on **one whole gene set** from one parent, the traits 'R' and 'y' of Fig. 8.5 would always stay **linked together** and could not be inherited independently.
- The answer: each gene set is not one long DNA thread but **separate independent pieces**, each called a **chromosome**.
- Each cell has **two copies of each chromosome**, one from the male parent and one from the female parent.
- Every germ cell takes **one chromosome from each pair**, which may be of **maternal or paternal** origin.
- When two germ cells combine, the **normal number of chromosomes is restored** in the progeny, ensuring the **stability of the DNA** of the species.

🤔 Think (NCERT asks this)

Asexually reproducing organisms also follow similar rules of inheritance. How? There is only **one parent**, so the offspring get a **copy of that parent's DNA**, with only small changes from copying errors. There is no mixing of two gene sets, so the offspring are nearly identical to the parent and variation builds up slowly.

8.2.4 Sex determination

How is the sex of a newborn individual decided? **Different species use very different strategies.**

🐢 Temperature

In a few **reptiles**, the temperature at which the fertilised eggs are kept decides whether the young will be male or female.

🐌 Change of sex

In **snails**, individuals can change sex, so sex is **not genetically determined**.

🧑 Genes

In **human beings**, sex is **largely genetically determined**: the genes inherited from our parents decide it.

Sex chromosomes in humans

- Most human chromosomes have a maternal and a paternal copy: we have **22 such pairs** (called autosomes).
- One pair, the **sex chromosomes**, is not always a perfect pair.
- **Women** have a perfect pair, both called **X**: women are **XX**.
- **Men** have a mismatched pair: one normal-sized **X** and one short **Y**: men are **XY**.
- So a human body cell has **23 pairs = 46 chromosomes** (22 pairs + XX or XY).

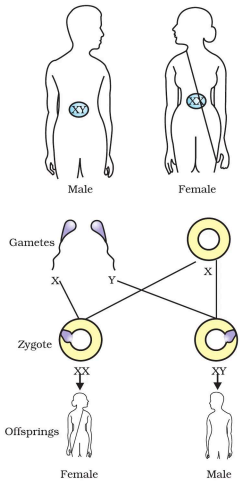


Fig. 8.6 Sex determination in human beings

Inheritance of X and Y (Fig. 8.6)

- Mother (XX) makes eggs that **all carry X**.
- Father (XY) makes two kinds of sperms in equal numbers: **half with X**, **half with Y**.
- Egg (X) + sperm (X) → **XX → girl**. Egg (X) + sperm (Y) → **XY → boy**.
- So **half the children will be boys and half girls** (on average), and the sex of the child is decided by **what it inherits from the father**.

Gametes	Sperm X (from father)	Sperm Y (from father)
Egg X (from mother)	XX: girl	XY: boy

⭐ Exam line

All children get an **X from the mother**. A child who gets an **X from the father** is a **girl**; one who gets a **Y from the father** is a **boy**. So the mother cannot be held responsible for the sex of the child; the chance of a boy or a girl is **50 : 50** in every pregnancy.

Differences to learn (2- and 3-mark favourites)

Dominant vs recessive trait

Dominant	Recessive
Expressed even when only **one copy** of its gene is present (TT or Tt)	Expressed only when **both copies** are recessive (tt)
Appears in the F1 generation	Hidden in F1; reappears in one-quarter of F2
Shown by a capital letter (T)	Shown by a small letter (t)
Example: tall plant, round seed, violet flower	Example: short plant, wrinkled seed, white flower

Genotype vs phenotype

Genotype	Phenotype
The genetic make-up (pair of genes) of an organism for a trait	The visible or observable trait
Cannot be seen directly; found from crosses	Can be seen
TT, Tt, tt	Tall, short
F2 monohybrid ratio 1 : 2 : 1	F2 monohybrid ratio 3 : 1

Monohybrid vs dihybrid cross

Monohybrid	Dihybrid
Parents differ in **one** pair of contrasting traits	Parents differ in **two** pairs of contrasting traits
Tall (TT) × short (tt)	Round green (RRyy) × wrinkled yellow (rrYY)
F2 phenotypic ratio **3 : 1**	F2 phenotypic ratio **9 : 3 : 3 : 1**
Shows dominance and that both traits pass to F1	Shows **independent inheritance** and new combinations

Homozygous vs heterozygous

Homozygous (pure)	Heterozygous (hybrid)
Both genes of the pair are the same: TT or tt	The two genes are different: Tt
Breeds true on self-pollination	Gives both types on self-pollination (3 : 1)
Makes one kind of gamete	Makes two kinds of gametes (T and t)

Autosomes vs sex chromosomes (humans)

Autosomes	Sex chromosomes
22 pairs	1 pair
Always a matched pair, same in males and females	XX in females (matched), XY in males (mismatched)
Carry genes for body characters	Decide the sex of the individual

Variation in asexual vs sexual reproduction

Asexual	Sexual
Very little variation	Much more variation
Only from small errors in DNA copying	From copying errors **and** mixing of DNA from two parents
Example: sugarcane field, bacteria	Example: human beings, pea plants

Previous year board questions

These questions come from the CBSE Class 10 Science board papers of the years shown, with the paper code (set) and marks. Tap a question to see a model answer.

Where we could not check a question word for word (for example its options or its case text), it is **reworded** and marked so. The years, sets and marks shown are only those we could confirm from the paper or a published solution of it.

2026 · 31/1/1 · 1 mark (reworded)

Which of the given statements about sex determination in human beings are correct? (Statements about the X-carrying gamete from the mother, the X-carrying gamete from the father and the Y-carrying gamete from the father.)

Know these facts to pick the right option: the **mother** gives only **X**-carrying eggs; the **father** gives either an **X**-carrying or a **Y**-carrying sperm, and it is the father's gamete that **decides the sex**. X from father → girl (XX); Y from father → boy (XY).

2026 · 31/1/1, Q15 (same question in set 31/1/3), case-based, 4 marks. Case (reworded): Mendel crossed a pure tall garden pea plant with a pure short one. All the F1 plants were tall.

2026 · 1 mark **(a) Why did only tall plants appear in the F1 progeny?**

The F1 plants are **Tt**. Tallness (T) is **dominant** over shortness (t), so a single T is enough to make the plant tall; the short trait is present but **not expressed**.

2026 · 1 mark **(b) What method did Mendel use to obtain the F2 progeny?**

He allowed the F1 tall plants to **self-pollinate** (F1 × F1).

2026 · 2 marks **(c) Write one difference between a dominant and a recessive trait.**

A **dominant** trait is expressed even when only one copy of its gene is present (TT or Tt), while a **recessive** trait is expressed only when both copies are recessive (tt). Example: tall (T) is dominant, short (t) is recessive.

2026 · 2 marks (OR) **(c) Write two observations made by Mendel about the F1 progeny.**

(i) **All** F1 plants were **tall**: only one parental trait was seen. (ii) There were **no halfway** (medium-height) plants: the traits did not blend.

2025 · 31/1/1 · 1 mark (options reworded)

Pea plants with round and green seeds (RRyy) are crossed with pea plants having wrinkled and yellow seeds (rrYY). What will the seeds of the F1 generation be like?

100% round and yellow (all RrYy). Round and yellow are the dominant traits; each F1 plant gets R and y from one parent and r and Y from the other.

– 2025 · 31/1/1 · 1 mark

Assertion (A): A human child bears all the basic features of human beings. Reason (R): It looks exactly like its parents, showing very little variations.

(c) A is true, but R is false. The child has the basic human design, but it does **not** look exactly like its parents: human populations show a great deal of variation.

– 2025 · 31/1/1 · 3 marks

(a) What are chromosomes? (b) Explain in brief how stability of DNA content of a species is ensured in sexually reproducing organisms.

(a) **Chromosomes** are thread-like structures in the nucleus that carry hereditary information (genes) in the form of **DNA**. Each gene set is present as separate pieces, the chromosomes.
(b) Each body cell has **two copies of each chromosome**, one from each parent. Germ cells are formed so that each takes **only one chromosome from each pair** (half the DNA). When a male and a female germ cell fuse at fertilisation, the **normal number is restored** in the zygote. So the DNA content stays the same from generation to generation.

– 2024 · 31/1/1 · 1 mark (reworded)

Which of the given statements about how the sex of a child is determined in humans are correct?

Correct facts: the sex is decided by the **father's sperm** (X sperm → girl, Y sperm → boy), and it is fixed **at fertilisation** when the gametes fuse. Statements saying the mother decides the sex, or that the mother gives a Y chromosome, are wrong.

– 2024 · 31/1/1 · 1 mark (reworded) **Which of the given statements about chromosomes are correct?**

Correct facts: chromosomes **carry hereditary information** (DNA), they appear as **thread-like structures in the nucleus**, and they take part in cell division. A statement that human **reproductive cells** (gametes) have chromosomes **in pairs** is wrong: gametes have only **one** chromosome of each pair.

2024 · 31/1/1, Q38, case-based, 4 marks (reworded). Case: Mendel's experiments with garden peas, crossing pure tall and pure short plants, and round-yellow (RRYY) with wrinkled-green (rryy) plants.

2024 · 1 mark

(i) State two observations Mendel made in the F1 plants of a cross between pure tall and pure short pea plants.

(i) All F1 plants were **tall** (the tall trait is dominant). (ii) There were **no intermediate** plants: the traits did **not blend**.

2024 · 1 mark **(ii) Give one difference between dominant and recessive traits.**

A dominant trait is expressed even with **one copy** of its gene; a recessive trait is expressed only when **both copies** are recessive.

2024 · 2 marks

(iii) In the cross RRYY × rryy, how was the F2 generation obtained, what ratio was seen and what can be concluded?

F2 was obtained by **self-pollination of the F1** (RrYy) plants. F2 ratio: round yellow : round green : wrinkled yellow : wrinkled green = **9 : 3 : 3 : 1**. Conclusion: new combinations appear, so the two traits (seed shape and seed colour) are **inherited independently**.

2024 · 2 marks (OR) **(iii) Justify: a trait may be inherited but not expressed.**

In a **Tt** plant, the recessive gene **t** is inherited but **masked** by the dominant T, so the plant is tall. The short trait is passed on (it reappears in F2 as tt) but is **not expressed** in the Tt plant.

2023 · 31/1/1, Q38, case-based, 4 marks (reworded). Case: Mendel's experiments with pea plants showing contrasting traits.

2023 · 31/1/1 **(a) Give examples of contrasting traits Mendel studied in pea plants.**

Tall / short (dwarf) plant; violet (purple) / white flower; round / wrinkled seed; yellow / green seed. (Any two pairs.)

2023 · 31/1/1 **(b) What are dominant and recessive traits?**

Dominant: the trait that is expressed even when only one copy of its gene is present. **Recessive**: the trait whose expression is masked by the dominant one; it shows only when two copies are present.

2023 · 31/1/1

(c) What phenotypic ratio is obtained in the F2 generation of a dihybrid cross, and what does it show? (OR: show the cross with a checkerboard.)

Ratio **9 : 3 : 3 : 1** (round yellow : round green : wrinkled yellow : wrinkled green). It shows that the two traits are **inherited independently** of each other. For the OR part, draw the 4 × 4 checkerboard with gametes RY, Ry, rY, ry, as in Fig. 8.5 (use the checkerboard tool to practise).

Practice

Board-pattern practice

Extra questions in the current CBSE pattern (not from a specific year). Tap to check.

Assertion–Reason (1 mark each)

(a) Both A and R are true, and R explains A. (b) Both true, but R does not explain A. (c) A true, R false. (d) A false, R true.

1 **A: In a cross between pure tall and pure short pea plants, all F1 plants are tall. R: Tallness is a dominant trait.**

(a)

2 **A: The sex of a child is determined by the mother. R: The mother has two X chromosomes.**

(d) The father decides the sex; R is true.

3 **A: New combinations of traits appear in the F2 of a dihybrid cross. R: The genes for seed shape and seed colour are inherited independently.**

(a)

4 **A: Germ cells have only one set of genes. R: Fertilisation restores the normal number of chromosomes.**

(a) Germ cells carry one set so that fusion gives back two sets.

5 **A: Variation is more in sexually reproducing organisms. R: Sexual reproduction involves DNA from only one parent.**

(c) It combines DNA from two parents.

Short answer

S1 **A tall pea plant is crossed with a short one. Half the offspring are tall and half are short. Write the genotypes of the parents.**

Tall parent **Tt**, short parent **tt**. Tt × tt gives Tt (tall) and tt (short) in a **1 : 1** ratio. (If the tall parent were TT, all offspring would be tall.)

S2 **How many types of gametes are made by a plant of genotype RrYy? Name them.**

Four: RY, Ry, rY, ry, in equal numbers.

S3 **Out of 800 F2 plants from a Tt × Tt cross, how many are expected to be (a) short, (b) pure tall?**

(a) Short (tt) = 1/4 × 800 = **200**. (b) Pure tall (TT) = 1/4 × 800 = **200** (the other 400 tall plants are Tt).

S4 **In a dihybrid F2 of 160 seeds, how many are expected to be wrinkled and green?**

1/16 × 160 = **10**.

S5 **Why is the probability of a boy or a girl 50 : 50 in every pregnancy?**

The father makes X sperms and Y sperms in **equal numbers**, and either one is equally likely to fertilise the egg (which always carries X). So XX (girl) and XY (boy) are equally likely.

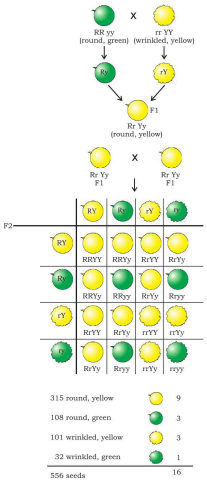
Case-based (4 marks)

C1 **A gardener crossed a pea plant with violet flowers with one having white flowers. All F1 plants had violet flowers. On selfing F1, he got 300 violet and 100 white plants. (i) Which trait is dominant? (ii) Write the genotypes of F1. (iii) Give the F2 genotypic ratio. (iv) How can he find out if a violet F2 plant is pure?**

(i) **Violet**. (ii) **Vv**. (iii) VV : Vv : vv = **1 : 2 : 1**. (iv) Let it self-pollinate (or cross it with a white vv plant): if all offspring are violet, it is **pure (VV)**; if some white appear, it is **Vv**.

Long answer (5 marks)

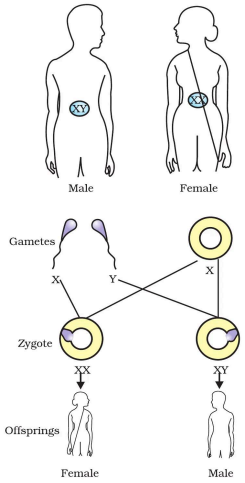
– **L1** **With the help of a cross, explain how Mendel's experiments show that traits are inherited independently. What ratio did he get?**



Model diagram: Fig. 8.5

P: RRyy (round green) × rrYY (wrinkled yellow); gametes Ry and rY. F1: RrYy, all round yellow (round and yellow dominant). F1 self-pollinated: each F1 plant makes four types of gametes, RY, Ry, rY, ry. The 16 combinations in the checkerboard give round yellow 9 : round green 3 : wrinkled yellow 3 : wrinkled green 1 (Mendel: 315 : 108 : 101 : 32). Combinations such as round-yellow and wrinkled-green, which were in neither parent, appear in F2. This is possible only if seed shape and seed colour are passed on **independently**, because each is on a separate chromosome and germ cells pick one chromosome of each pair at random.

– **L2** **Explain with a flow chart how the sex of a child is determined in human beings. Why is it wrong to blame the mother for the birth of a girl?**



Model diagram: Fig. 8.6

Father (44 + XY) makes sperms 22 + X and 22 + Y in equal numbers; mother (44 + XX) makes eggs that are all 22 + X. Egg X + sperm X → 44 + XX → girl. Egg X + sperm Y → 44 + XY → boy. The mother always gives X, so the sex depends on **which sperm** fertilises the egg, which is a matter of chance. The mother has no role in deciding it.

Recap

What you have learnt

- Variations arising during reproduction can be **inherited**; they **accumulate** over generations. Sexual reproduction gives more variation than asexual.
- These variations may lead to **increased survival** of individuals (heat-resistant bacteria in a heat wave). Selection of variants by the environment is the basis of evolution.
- Sexually reproducing individuals have **two copies of genes** for the same trait. If the copies are not identical, the trait that is expressed is **dominant**, the other **recessive**.
- Monohybrid cross: F1 all tall; F2 **3 : 1** (phenotype), **1 : 2 : 1** (genotype).
- Traits may be **inherited separately**, giving new combinations in the offspring (dihybrid F2 **9 : 3 : 3 : 1**).
- A **gene** is a section of DNA that codes for one protein; genes control traits (gene → enzyme → hormone → height).
- Genes are on separate **chromosomes**; each germ cell takes one of each pair; fertilisation restores the number and keeps DNA stable.
- Sex is decided by different factors in various species (temperature in some reptiles, sex change in snails). In humans: 22 pairs + XX (girl) or XY (boy); the **father's** chromosome (X or Y) decides.

Quiz Test yourself

Tap an answer. **Score: 0 / 15**

1. Tall violet × short white gives all violet, about half short. The tall parent is (NCERT Ex. 1)

- TTWW
- TTww
- **TtWW**
- TtWw

Half short → Tt; all violet → WW.

2. The F2 phenotypic ratio of a monohybrid cross is

- 1 : 2 : 1
- **3 : 1**
- 9 : 3 : 3 : 1
- 1 : 1

1 : 2 : 1 is the genotypic ratio.

3. The F2 genotypic ratio of a monohybrid cross is

- **1 : 2 : 1**
- 3 : 1
- 2 : 1
- 1 : 1 : 1 : 1

TT : Tt : tt.

4. RRyy × rrYY. The F1 seeds are

- Round, green
- Wrinkled, yellow
- **Round, yellow**
- Wrinkled, green

All RrYy: round and yellow are dominant (CBSE 2025).

5. In a dihybrid F2 of 16, the number with both recessive traits is

9	3
1	0

Only rryy (wrinkled green).

6. A section of DNA that provides information for one protein is a

Chromosome	**Gene**
Nucleus	Trait

NCERT definition of a gene.

7. Which plant is heterozygous?

TT	tt
Tt	TTYY

Two different copies of the gene.

8. Human females have sex chromosomes

XX	XY
YY	XO

Men are XY.

9. A child who gets a Y chromosome from the father will be a

- **Boy**
- Girl
- Either
- Depends on the mother

X (mother) + Y (father) = XY.

10. Number of pairs of autosomes in humans

- 23
- **22**
- 46
- 44

22 pairs + 1 pair of sex chromosomes = 23 pairs.

11. In some reptiles, sex is decided by

- X and Y chromosomes
- **Temperature of the eggs**
- The mother's diet
- Changing sex

Snails change sex; humans use X and Y.

12. Mendel's F1 from tall × short had

- All short plants
- Medium-height plants
- **All tall plants**
- Tall and short in 3 : 1

No halfway plants; tall is dominant.

13. Mendel got the F2 generation by

- Crossing F1 with the short parent
- **Self-pollinating F1 plants**
- Vegetative propagation
- Crossing two parents again

CBSE 2026 asked this.

14. Tt × tt gives tall and short in the ratio

- 3 : 1
- **1 : 1**
- 1 : 2 : 1
- All tall

Half Tt, half tt.

15. A field of sugarcane shows very little variation because sugarcane

- Has no DNA
- **Is grown asexually (vegetatively)**
- Self-pollinates
- Has only recessive traits

Asexual reproduction gives only minor variations.

NCERT Answers to all questions

Every question from the chapter, with answers you can write in the exam. Tap to open.

Questions (page 129)

1. If a trait A exists in 10% of a population of an asexually reproducing species and a trait B exists in 60% of the same population, which trait is likely to have arisen earlier?

Trait B. In asexual reproduction, a new trait arises from a small error in DNA copying in one individual and is then passed on to all its descendants. The longer ago a trait arose, the more generations it has had to spread, so it is found in a larger share of the population. Trait B (60%) is therefore likely to be older than trait A (10%).

2. How does the creation of variations in a species promote survival?

Variations give different individuals different advantages. If the environment changes (for example a heat wave), individuals with a suitable variation (heat-resistant bacteria) survive and reproduce, while others die. So the species as a whole is not wiped out. Without variation, one drastic change could kill the whole population.

Questions (page 133)

1. How do Mendel's experiments show that traits may be dominant or recessive?

Mendel crossed pure tall (TT) and pure short (tt) pea plants. All F1 plants were **tall**: only one trait (tallness) was expressed, though the F1 plants had received a factor for shortness too. When F1 plants were self-pollinated, **one-quarter** of the F2 plants were **short**. So the shortness trait had been inherited by F1 but stayed hidden. The trait that shows in F1 (tall) is **dominant**; the one that is hidden in F1 and reappears in F2 (short) is **recessive**.

2. How do Mendel's experiments show that traits are inherited independently?

Mendel crossed plants differing in two traits, e.g. tall with round seeds × short with wrinkled seeds (or RRyy × rrYY). F1 plants all showed the two dominant traits. In F2, besides the parental combinations, **new combinations** appeared: tall plants with wrinkled seeds and short plants with round seeds, in the ratio 9 : 3 : 3 : 1. If the two traits were passed on together, these new combinations could not appear. So the traits are inherited **independently**.

3. A man with blood group A marries a woman with blood group O and their daughter has blood group O. Is this information enough to tell you which of the traits – blood group A or O – is dominant? Why or why not?

No. The result fits both possibilities. If A were dominant, the father could carry one A and one O copy and pass on O, giving an O daughter. If O were dominant, the daughter could simply have got the dominant O from the mother. One family with one child is not enough: we need data from **many crosses** and their ratios over generations to decide which trait is dominant.

4. How is the sex of the child determined in human beings?

Women have a matched pair of sex chromosomes (XX) and men a mismatched pair (XY). All eggs carry an **X**. Half the sperms carry **X** and half carry **Y**. If an X sperm fertilises the egg, the child is XX, a **girl**; if a Y sperm fertilises it, the child is XY, a **boy**. So the sex of the child is decided by the chromosome inherited from the **father**.

Activities

– **8.1 Earlobe survey: suggest a possible rule for the inheritance of earlobe types.**

See the answer in section 8.2.1: free earlobe behaves as dominant and attached as recessive; a child is attached-lobed only when both copies are "attached".

– **8.2 What experiment would confirm the 1 : 2 : 1 ratio of TT, Tt and tt in F2?**

Self-pollinate every F2 plant and grow the next generation: tt gives only short, TT (1/3 of the tall plants) gives only tall, Tt (2/3 of the tall plants) gives tall and short in 3 : 1. See section 8.2.2.

– **Fig. 8.4 Work out which trait would be considered dominant and which one recessive in Fig. 8.4.**

Violet flower colour is dominant; **white** is recessive (F1 all violet, F2 3 violet : 1 white).

Exercises (page 133)

– **1. A Mendelian experiment consisted of breeding tall pea plants bearing violet flowers with short pea plants bearing white flowers. The progeny all bore violet flowers, but almost half of them were short. This suggests that the genetic make-up of the tall parent can be depicted as (a) TTWW (b) TTww (c) TtWW (d) TtWw**

(c) TtWW. Almost half short → the tall parent is Tt (Tt × tt gives 1 tall : 1 short). All violet → the parent is WW (if it were Ww, about half would be white).

– **2. A study found that children with light-coloured eyes are likely to have parents with light-coloured eyes. On this basis, can we say anything about whether the light eye colour trait is dominant or recessive? Why or why not?**

No. Light-eyed parents having light-eyed children only shows that the trait is inherited. It fits both cases: if light is recessive, two light-eyed (recessive) parents can only have light-eyed children; if light is dominant, light-eyed parents could also have light-eyed children. To decide, we would need to study crosses between light-eyed and dark-eyed parents and count the ratio of the two types in their children and grandchildren.

3. Outline a project which aims to find the dominant coat colour in dogs.

1. Choose two **pure-breeding** dogs with contrasting coat colours, e.g. black (BB) and white (bb), whose ancestors always had the same colour.
2. Cross them and observe the puppies (F1). The colour shown by **all F1 puppies** is the **dominant** one; the hidden colour is recessive.
3. To confirm, cross F1 dogs with each other. In F2, the dominant and recessive colours should appear in about **3 : 1**.
4. Repeat with several litters to get large numbers and reliable ratios.

4. How is the equal genetic contribution of male and female parents ensured in the progeny?

Each body cell has **two copies of each chromosome**, one from each parent. When germ cells (sperm and egg) are made, each gets **only one chromosome from every pair**, i.e. half the usual number. At fertilisation, one male germ cell fuses with one female germ cell, so the zygote gets **half its chromosomes from the father and half from the mother**, and the normal number is restored. In humans, each parent gives 23 chromosomes (22 + X from the mother; 22 + X or Y from the father).

Diagrams from the NCERT Science textbook for Class 10 (reprint 2026-27). Board questions from CBSE Class 10 Science papers 2023 to 2026.